

Trajectory and Force

If you've ever thrown a ball, you know that the harder you throw, the farther it travels. But the force of the throw isn't the only thing that affects the ball's distance. The angle at which the ball is thrown will also affect its **trajectory**, or path through the air.

Newton's first law of motion says that an object in motion stays in motion until another force acts on it. That other force may be a bat, a racket, a paddle, a club, a player, or another ball, but at some point, the other force is **gravitational force**.

Everything that goes up must come down because of gravity. If a ball goes straight up, exactly 90 degrees from the ground, it will land in exactly the same spot, no matter how much force propelled the ball into the air. If you throw it forward at a lower angle, it will travel in an **arc**, or curve, some distance before it lands. The distance depends on the **angle** and the **amount of force** used. In general, a ball thrown at a high angle doesn't travel as far as it does at a lower angle. But if the angle is too low, the ball drops to the ground too quickly to go very far.

It is difficult for a person to throw a ball exactly the same way every time, but a launching device is much more consistent. A launcher often uses a flat, stiff object called a **lever** and a **fulcrum**, a pivot that the lever pushes against. The lever sends a ball flying when the end opposite from the ball has a force applied to it. If the launcher uses a spring or stretchy material containing **potential energy** to hold back the end with the ball, the ball moves when the spring is released. The length of the lever and the height and position of the fulcrum, along with the amount of force used, affect the trajectory of the ball.

STEM Challenge: Ball Tosser

Visual Literacy

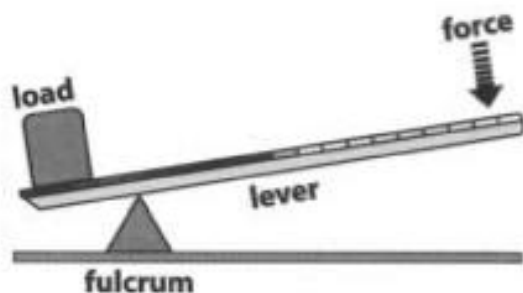
Angles of a circle

A circle can be divided into 360 units called **degrees**. Starting from the flat, horizontal ground at 0 degrees, 90 degrees is straight up in the air.



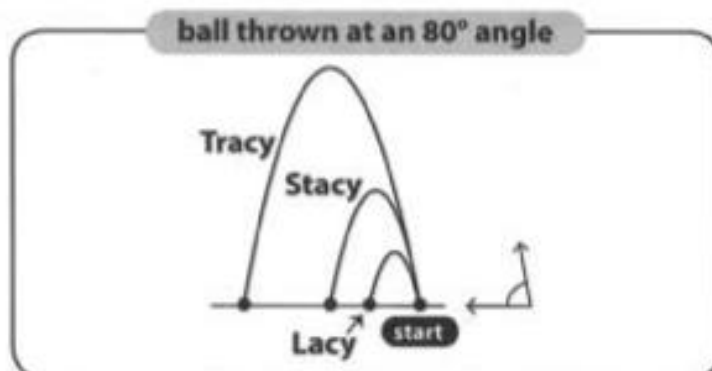
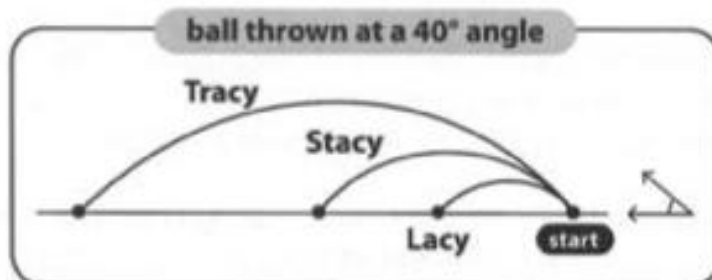
Lifting a load

The fulcrum can be placed anywhere along the lever. Its position, along with its height and the amount of force used, affects what happens to the load.



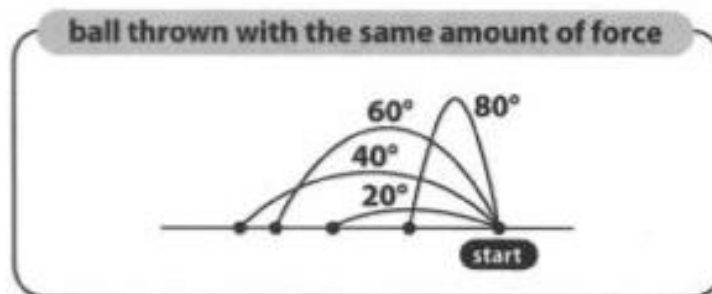
Comparing different amounts of force

Lacy, Stacy, and Tracy all threw the ball at a 40-degree angle and again at an 80-degree angle, but they each used different amounts of force. Lacy threw it with the least force; Tracy used the most force.



Comparing different angles

Stacy threw the ball at 80 degrees, 60 degrees, 40 degrees, and 20 degrees, all using the same amount of force.



STEM Challenge:

Ball Tosser

Challenge

Ball Tosser

Challenge: Design a machine that tosses a ball into the air and into a container.

Testable goal: The ball reaches a container 10 feet (3 meters) away.

Research: Look at pictures of catapults and other launching machines. Notice how the machines are constructed. Think about how science, technology, engineering, and math are used to create these machines.

Brainstorm: Draw one or more design ideas for a ball tosser. There are many different ways to complete this challenge. Be creative!

A large, empty rectangular box with a double-line border, intended for students to draw their design ideas for a ball tosser.

STEM Challenge: Ball Tosser

Suggested Materials List

Item for each group

- ☐ scissors

Items for the whole class

- | | |
|---|--|
| ☐ yard/meter sticks | ☐ rubber bands or elastic |
| ☐ foot rulers | ☐ springs |
| ☐ handheld strainers | ☐ _____ |
| ☐ round cup or pint containers | ☐ _____ |
| ☐ soup ladles or wooden spoons | ☐ _____ |
| ☐ cans, different sizes | ☐ _____ |
| ☐ doorstops | ☐ _____ |
| ☐ cardboard or tagboard | ☐ _____ |
| ☐ shoe boxes | ☐ _____ |
| ☐ duct or masking tape | ☐ _____ |

Items for testing

- | | |
|---|---------------------------------------|
| ☐ foam or hollow ball | ☐ tape measure |
| ☐ basket, bucket, or trash can | |

STEM Challenge: Ball Tosser

Think About the Design Process

1 Plan

Think about the materials available to you. How can you use the materials to create a ball tosser?



2 Create

Use the materials you have to create a ball tosser. Follow your design as closely as possible.



3 Test

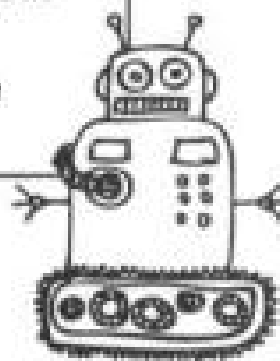
Use your ball tosser. Does it toss a ball? Can you control the toss and make the ball go into a basket 10 feet (3 meters) away?



4 Evaluate and Revise

Evaluate the performance of your ball tosser. What revisions can you make to improve its performance?

Repeat the design process until you are satisfied with your ball tosser.



STEM Challenge: Ball Tosser

Design Process

1 Plan: Describe and draw your design. List your materials.



2 Create



3 Test: Describe and draw your results.



4 Evaluate

STEM Challenge: Ball Tosser

Redesign Process

1 Redesign: Describe what you will change. Draw your new design.



2 Revise or re-create



3 Test: Describe and draw your results.



4 Evaluate